

Production Capstone - Marketplace Data Platform

Protected final build | Requirements only | RC2

Business brief

A multi-region marketplace needs a trustworthy data platform for orders, refunds, seller operations and delivery events. Sources include a paginated API, daily CSV drops and a PostgreSQL operational database. Analysts need daily Gold tables; operations needs a near-real-time delivery exception view. The platform must survive reruns, late data, schema changes and partial failures.

Constraints

- Preserve raw source evidence.
- No plaintext secrets.
- Use explicit grain/business keys and reconcile critical counts/totals.
- Use an incremental strategy where appropriate.
- Orchestrate the repeatable workflow.
- Use distributed processing only where justified.
- Use Microsoft Fabric as the primary cloud deployment target.
- Streaming is optional unless you justify it for the delivery-exception requirement.

Required deliverables

- Architecture and source contracts
- Ingestion code and raw landing evidence
- Quality/quarantine/reconciliation
- Trusted relational/lakehouse transformations
- Orchestration and rerun/backfill procedure
- Testing/CI/security/monitoring evidence
- README, runbook, data dictionary and known limitations
- Technical defense covering trade-offs, recovery and 10x scale

Protected rule

The capstone intentionally provides no completed model, DAG, Spark job or architecture answer. Save your first implementation before opening the incident pack.